Vahab Jabrayilov

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vjabrayilov.github.io

In US with work authorization (CPT/OPT)

Education

2023 - 2027 (Expected) **Ph.D., Computer Science**, *Columbia University*, New York, NY, USA
Advisor: Prof. Kostis Kaffes
Interests: Low Latency Systems Optimization, Scheduling, Networking, ML Systems Performance

2023 - 2025 **M.Sc., Computer Science**, *Columbia University*, New York, NY, USA

2019 - 2023 **B.Sc., Computer Engineering**, *Middle East Technical University*, Ankara, Turkey
GPA: 3.8/4.0 Ranking: 7/305

Work Experience

Aug 2023 - Current **Columbia University**, *Graduate Research Assistant*, New York, NY, USA
- Built Machnet(C++/DPDK), a kernel-bypass networking library with $< 100\mu s$ tail latency, accelerating distributed training and data pipelines.
- Designed and implemented a microsecond-scale Virtual Machine scheduler (Rust/ sched_ext/eBPF), enabling fine-grained performance control for latency-sensitive ML workloads.
- Removed nanosecond-scale jitter from scheduling/networking data paths via perf, bpftrace and hardware counter profiling.

Jun-Sep 2024 **Microsoft Research**, *Research Software Engineer Intern*, Redmond, WA, USA
- Mentors: Srikanth Kandula, Sathiya Kumaran Mani @ Networking Research Group
- Developed Rust/DPDK based packet processor for Azure's real-time data ingestion pipelines, processing millions of flow logs/second with deterministic, sub-millisecond latency.
- Built stress-testing framework to ensure reliability of *Zero Trust Networking* service of Azure.

Jun-Aug 2023 **EPFL**, *Research Intern*, Lausanne, Vaud, Switzerland
- Advisor: Prof. Sanidhya Kashyap @ RS3LAB
- Analyzed and designed mitigations for metastable failures in distributed systems to improve uptime in high-availability systems, e.g. large scale ML clusters.

Publications

Under Review, 2025 Rorke: Black-box Microsecond-scale VM Scheduling
Vahab Jabrayilov, Teng Jian, Jason Nieh, Kostis Kaffes

Under Review & arXiv, 2025 Fast Userspace Networking for the Rest of Us
Alireza Sanaee*, **Vahab Jabrayilov***, Ilias Marinos, Anuj Kalia, Divyanshu Saxena, Prateesh Goyal, Kostis Kaffes, Gianni Antichi (* equal contribution)

VLDB, 2025 HoliPaxos: Towards More Predictable Performance in State Machine Replication
Zhiying Liang, **Vahab Jabrayilov**, Aleksey Charapko, Abutalib Aghayev

ACM SoCC, 2024 Rethinking the Networking Stack for Serverless Environments: A Sidecar Approach
Vishwanath Seshagiri, Abhinav Gupta, **Vahab Jabrayilov**, Avani Wildani, Kostis Kaffes

Projects

GitHub Repo **Machnet: Ultra-Low Latency Networking Library**
Ultra-low latency networking library tuned for high-frequency low-latency communication systems, e.g. distributed ML data exchange; delivered $750k$ req/sec with $61\mu s$ tail latency on commodity cloud.

GitHub Repo **Replicant: Strongly Consistent Key-Value Store**
MultiPaxos-based key-value store with deterministic replication, designed for consistent state management in fault-tolerant infrastructure like consistent parameter synchronization in replicated ML services.

Skills

Languages/Platforms Assembly(x86, Arm), C/C++, CUDA, PyTorch, TensorFlow, Go, Java, Python, Rust, SQL

Tools Abseil, AWS, Azure, Boost, Docker, DPDK, eBPF, Git, Kubernetes, RDMA, TLA+, Tokio